Explore Linear Time-Invariant Systems

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Use the `starter.m` file to complete the following exercises. Be sure to change the placeholder lines of the starter file to match specific information for this assignment.

The file `CarefreeA.wav` contains a recording sampled at 44,100 samples per second. The starter code loads this file into the MATLAB workspace and plays it for you. Note that you can hit any key with the cursor in the command window to resume the code following the `pause` command. The pause is necessary to prevent later sounds from playing simultaneously with the previous ones.
We will study the following two systems described by the difference equations.
System 1:

$$y[n] - 2.6862y[n-1] + 2.4197y[n-2] - 0.7302y[n-3] = 10^{-3}\{0.4165x[n] + 1.2496x[n-1]$$
$$+1.2496x[n-2] + 0.4165x[n-3]\}$$

System 2:

$$y[n] - 1.7600y[n-1] + 1.1829y[n-2] - 0.2781y[n-3] = 0.5276x[n] - 1.5829x[n-1]$$
$$+1.5829x[n-2] - 0.5276x[n-3]$$

1. You may apply each of these systems to the music piece using the filter command as illustrated in the starter file.

Use soundsc to listen to both versions of the sound. Based on what you heard, which system passes low frequencies, and which system passes high frequencies?

2. a) Use the filter command to find and display (using stem) the first 120 values of the impulse response of each system. You may assume the systems are causal - that is, the impulse response is zero prior to time zero. Example code is in the starter file.

b) Now use freqz to find the frequency response of each system and plot the magnitude and phase as illustrated in the starter file.

Does the frequency response magnitude confirm your conclusion about which systems pass low and high frequencies? Explain.

3. a) Use the filter command to pass the low- and high-frequency sinusoidal signals defined in the starter file through each system.

Use stem to display the result. How does each system respond to each input? Is this predicted by the frequency response of the systems? Does the gain experienced by each cosine agree with the gain in the frequency response at each frequency? Note that these are not true sinusoids because they start at time zero - filter assumes the input is zero prior to the start of the signal - so there is a start up phenomenon. The apparent length of the impulse response is a good indicator of how long the start up effects can last.

b) Now use the impulse response and convolution to compute the output of the first system for each input using the conv command as illustrated in the starter file.

Use stem to display the first 250 samples of the result. How does the result obtained using convolution relate to that obtained using the difference equation?